

Nikolaos Liakas

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Summary

A recent MSc (Dist) Computer Science graduate, transitioning after five years in data engineering in financial services to software engineer. I have a unique blend of both more than five years experience in the hands-on design, planning, implementation and testing of software, as well as the academic and research background of infrastructure aware development. My interests are compiler design, hacking memory allocators and static analysis. This is combined with my [projects](#) in C++ and OCaml.

Projects in Development

Mallocopter

Dynamic memory allocation is a fundamental service expected of every operating system. This repository models the process and strategies used from simple bump-allocations to full featured allocators like `ptmalloc`. The project is in C++. This is modelled through a simple array that demonstrates the various strategies used in a modern allocator for memory management of user-processes in virtual memory. The long-term aim would be to use this project to test custom allocators for program or domain specific requirements.

GB-Emulator

The original Gameboy had one of the most elegant memory models in device history. This repository is a gameboy emulator written in C++. It takes real ROMs and allows the computer user to play games!

Work Experience

Lead Analytics Engineer, Data and Analytics

Liberty Specialty Markets (LSM) | April 2023 - Present

- Leading the design, planning, implementation and testing of data applications within LSMs Analytics team.
- Primary coding languages are Python and SQL
- The data applications range from full featured microservices with REST API interfaces, to small AWS Lambda and Glue stateless services orchestrated in Airflow
- Code structured into reusable python packages with OOP patterns where suitable
- CICD is provided in GitHub Actions - strong and quick test-suites provided across data applications
- Implemented Architectural Design Records and Data Contracts for standardisation and observability of data applications across the estate
- RDBMSs include: MS SQL Server, Redshift and PostgreSQL

Senior Consultant, Data Engineering and Architecture

EY | August 2021 - April 2023

- Technical programming lead on a credit-risk model development engagement for a UK/APAC bank: translated legacy SAS/SQL models to Python and PySpark; used NumPy and SciPy for numerical computation
- Technical lead for an internal physical-risk loss-modelling framework: engineered the application architecture and led a three-person team (two developers, one quant). Sole developer of probability of default models
- Managed data-change testing workflow for a large American bank, validating system behaviour under issue-management-driven code changes
- Git management and design for large team - training given for CICD, TDD and feature/trunk -based branch strategies

AVP, Product Data Lead

Barclays UK | March 2018 - July 2021

- Product Data Lead on a high-value regulatory review project spanning interest-rate derivatives, pensions and retail banking products
- Designed and implemented a relational data model in SQL Server/Access; defined and automated program orchestration using Airflow

- Extensive Python program design with NumPy and pandas; built a VBA userform for structured team inputs. Managed a team of four including budget governance for two contractors

Education

Oct 2023 – December 2025	Birkbeck College, University of London – London, UK MSc Computer Science - Distinction Thesis: LCSTRS of OCAML
Feb 2015 – May 2016	École nationale d'horlogerie du Canada – Trois-Rivières, QC, Canada DEP Horlogerie – Diploma in Horology
Sept 2013 – Sept 2014	Goldsmiths College, University of London – London, UK MA Contemporary Art Theory - Merit
Sept 2009 – June 2013	University of Toronto, St. George Campus – Toronto, ON, Canada (Hons.) BA Majors: Italian Lang. and Literature and Art History Minor: Bosnian, Croatian, Serbian Lang. and Literature

References

Available upon request.